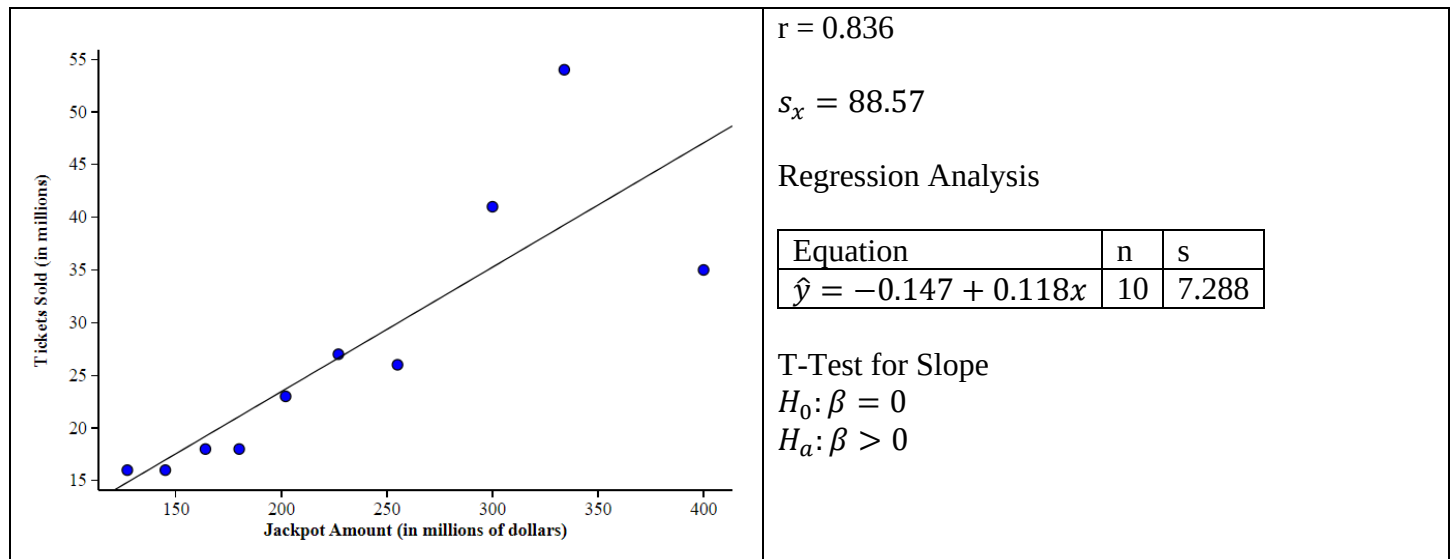


HW 2

1) The largest winning jackpot in Mega Millions history was \$1.537 billion, for the October 23, 2018, drawing, in which a single winning jackpot ticket was sold in South Carolina. Ticket sales leading up to the Jackpot being won were unprecedented. A statistician was interested in determining if the number of tickets sold for a Mega Millions drawing depended on how large the Jackpot was. They randomly selected 10 Mega Millions drawings in history and recorded how many ticket sales there were for that drawing. The data below shows the result of the investigation.



(a) What is the standard error of the slope of the regression line? Interpret it in the context of the problem.

(b) What do you know (given the information about) or what would you need to know in order to perform T-Test for the slope of the regression line?

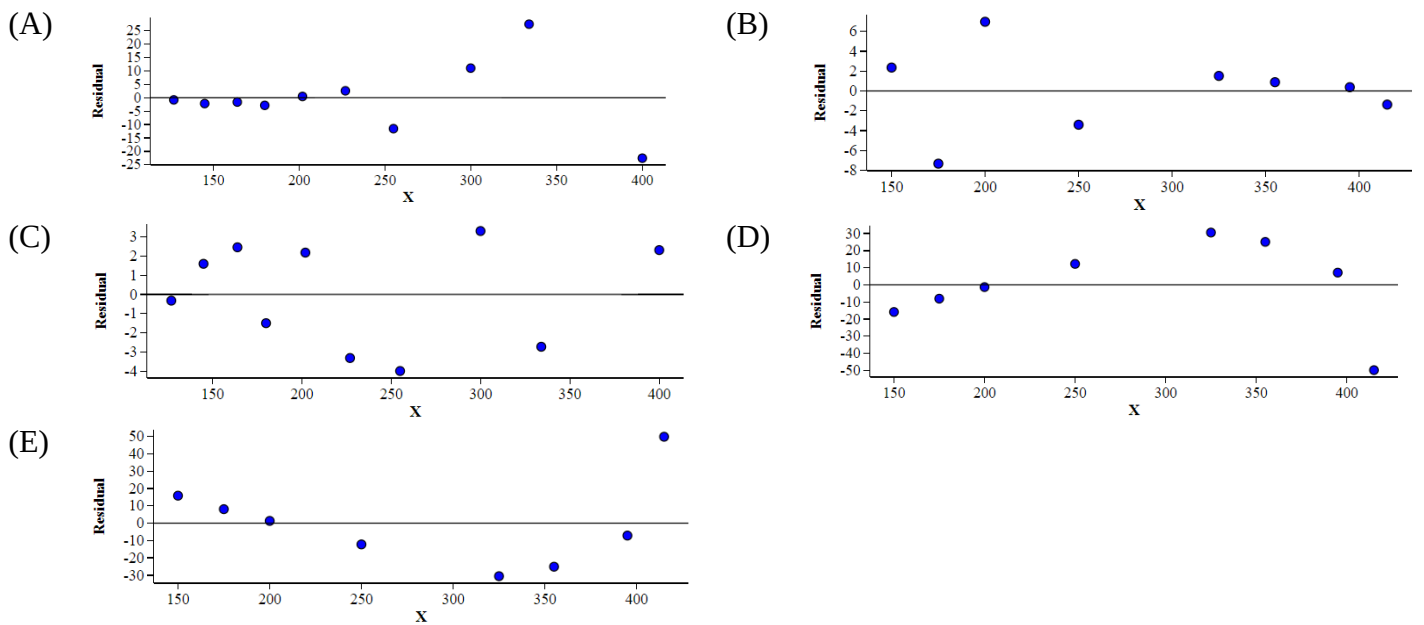
(c) Assuming that the conditions needed for doing inference for regression are present, what is the correct t-test statistic for the listed hypothesis?

(d) Find and interpret the p-value with the test statistic you calculated in part (c). What is the conclusion to this test in the context of the problem? Use a significance level of 5%.

_____ 2) If a test of hypotheses rejects $H_0: \beta = 0$ in favor of the alternative hypothesis $H_a: \beta > 0$, where β is the population regression slope, then the least-squares regression line

- (A) is useful for predicting y , given x (within the limits of x -values covered by the data).
- (B) slopes downward and to the right when plotted on the scatterplot of paired observations (x, y) .
- (C) can be extrapolated beyond the limits of the x -values covered by the data to predict y at any possible x .
- (D) is not useful for predicting y , given x .
- (E) has an intercept that is greater than zero.

_____ 3) Which of the following residual plots correctly shows the standard deviation of the residuals remaining approximately constant?



Use the scenario below for questions 4 to 6:

Mrs. Maslowski wonders if she can predict the word count of the essays her students submit, based on how large the file size is. She selects a random sample of 13 student papers and compares file size (measured in kilobytes) to the word count for each paper. A regression analysis of her data is given below. Assume all conditions for inference for slope have been met.

Predictor	Coef	SE Coef	T	P
Constant	-1450.3	773.4	-1.88	0.088
File Size	76.50	28.60	2.67	0.022
S = 145.851 R-Sq = 39.4% R-sq (adj) = 33.9%				

_____ 4) Which of the following is the estimate of the standard deviation of the sampling distribution of b ?

- (A) 773.4 (B) 145.851 (C) $\frac{773.4}{\sqrt{13}}$ (D) 28.60 (E) $\frac{28.60}{\sqrt{13}}$

_____ 5) Which of the following is the best interpretation of $s = 145.851$?

- (A) The average distance between the file size of each of these essays and the mean file size for all the essays in this sample is about 146 KB.
(B) The sum of the squared deviations between each observed file size and the file size predicted by the regression equation is 146 KB.
(C) The average distance between the words counts of each of these essays and the mean word count for all the essays in this sample is about 146 KB.
(D) The sum of the squared deviations between each observed word count and the word count predicted by the regression equation is 146 KB.
(E) Predictions of word count from file size based on this regression model will be off by an average of about 146 KB.

_____ 6) If Mrs. Maslowski uses these data to test the hypotheses $H_0: \beta = 0$ vs $H_a: \beta \neq 0$, at the $\alpha = 0.05$ level, which of the following is an appropriate conclusion?

- (A) Since the P-value of 0.022 is less than α , reject H_0 . We have convincing evidence that there is a linear relationship between file size and word count.
(B) Since the P-value of 0.022 is less than α , fail to reject H_0 . We do not have convincing evidence that there is a linear relationship between file size and word count.
(C) Since the P-value of 0.088 is greater than α , reject H_0 . We have convincing evidence that there is a linear relationship between file size and word count.
(D) Since the P-value of 0.088 is greater than α , fail to reject H_0 . We do not have convincing evidence that there is a linear relationship between file size and word count.
(E) Since the P-value of 0.088 is greater than α , accept H_0 . We have convincing evidence that there is a linear relationship between file size and word count.